

Day 21 - Best Candidate Review : SVM

Baseline vs Tuned Results

Model	Version	Accuracy	Precision	Recall	F1	BER
Logistic Regression	Baseline	74.17%	66.95%	61.70%	65.17%	0.258
Logistic Regression	Tuned (C=0.1, lbfgs)	74.17%	66.95%	61.70%	65.17%	0.258
SVM	Baseline	75.83%	84.62%	46.81%	60.27%	0.242
SVM	Tuned (C=0.1, Linear)	75.83%	84.62%	46.81%	60.27%	0.242

Observation :

- Hyperparameter tuning did not significantly improve test-set performance.
- Baseline models were already close to optimal.
- SVM remains the strongest overall model.
- Logistic Regression remains a strong secondary candidate.



Representative Successes :



Success Case 1

- Low NTU condition.
- Clear separation between bit 0 and bit 1.
- Both models correctly classified all bits.

Success Case 2

- Medium NTU condition.
- Small signal variations.
- Models maintained high accuracy (>89%).

Representative Failures :



Failure Case 1

- High NTU condition.
- Significant signal attenuation.
- Bit 0 and bit 1 values overlap.

Failure Case 2

- Samples near decision boundary.
- Model predicts incorrect bit.

Reson for Failure

- Small dataset size (399 samples).
- Limited feature set (Analog Value, Analog Voltage, NTU).
- Performance degrades under high turbidity conditions.
- No real-time underwater testing available.